NETFORGE

Network Security Testing and Assessment Platform

Project Report

Field	Details
Project Title	NETFORGE — Network Security Testing and Assessment Platform
Platform	Kali Linux
Technology	Python, Flask, SQLite, HTML/CSS/JS, Nmap, TShark, Traceroute
Project Type	Cybersecurity / Network Security / Security Assessment
Repository	GitHub – NETFORGE

1. INTRODUCTION

NETFORGE is a Kali Linux-based network security testing and assessment platform designed to provide a centralized interface for performing authorized network analysis and security assessments.

The project combines multiple networking and security utilities into a single web-based platform. It provides functionality for network scanning, DNS analysis, traceroute analysis, traffic monitoring, security-risk assessment, scan history, and automated HTML report generation.

NETFORGE was developed to make common network-security testing activities easier to perform and understand through a professional dashboard rather than requiring every operation to be performed manually from the command line.

2. PROBLEM STATEMENT

Network security testing often requires the use of multiple independent command-line utilities such as Nmap, TShark, DNS utilities, and traceroute.

For beginners and security professionals, manually executing these tools and interpreting their output can be time-consuming. It can also make it difficult to maintain historical records and generate structured reports.

Therefore, NETFORGE was developed to provide a unified platform that:

- Performs authorized network discovery.
- Collects network information.
- Analyzes discovered services.
- Captures and analyzes network traffic.
- Identifies potential security risks.
- Maintains scan history.
- Generates structured security reports.
- Provides the results through a web dashboard.

3. OBJECTIVES

The primary objectives of NETFORGE are:

1. To develop a centralized network-security testing platform.
2. To integrate commonly used Kali Linux networking tools.
3. To automate network information collection.
4. To analyze discovered services and potential risks.
5. To provide a risk-based security assessment.
6. To monitor and analyze network traffic.
7. To store scan results for historical analysis.
8. To generate professional HTML security reports.

9. To provide an easy-to-use web interface.
10. To support authorized security testing in controlled environments.

4. SCOPE OF THE PROJECT

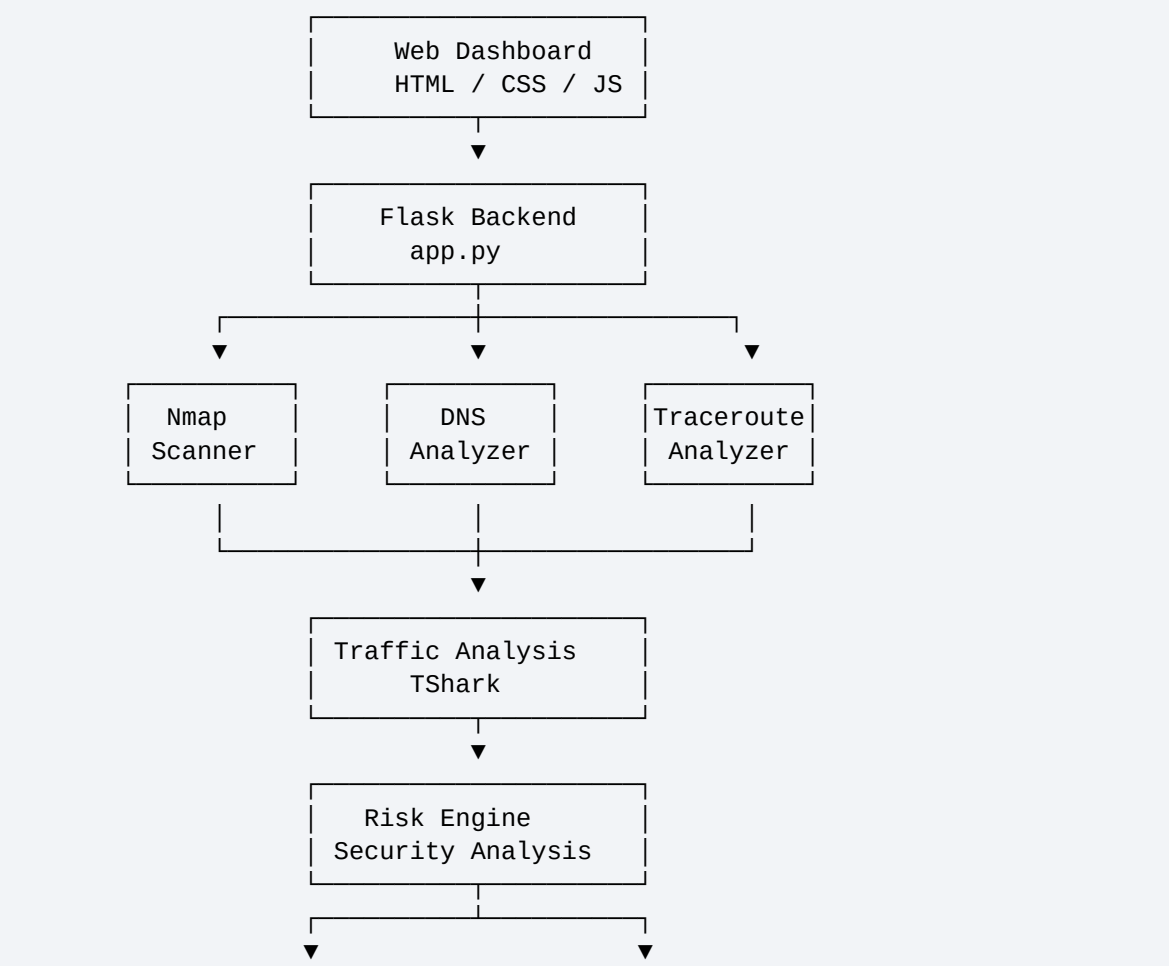
NETFORGE is designed primarily for:

- Cybersecurity learning.
- Network-security laboratories.
- Authorized penetration-testing environments.
- Security assessment of systems owned by the tester.
- Internal network testing.
- Educational demonstrations.
- Security documentation and reporting.

The platform is not intended for unauthorized scanning or monitoring of systems or networks.

5. SYSTEM ARCHITECTURE

The overall architecture of NETFORGE is:





6. TECHNOLOGIES USED

Category	Technology
Operating System	Kali Linux
Programming Language	Python
Backend	Flask
Frontend	HTML, CSS, JavaScript, Chart.js
Database	SQLite
Security & Network Tools	Nmap, TShark, Traceroute
Development Tools	Git, GitHub, Visual Studio Code / Kali terminal

7. PROJECT STRUCTURE

```
NETFORGE/
│
├── backend/
│   ├── app.py
│   ├── scanner.py
│   ├── network.py
│   ├── dns.py
│   ├── traffic.py
│   ├── analyzer.py
│   ├── risk_engine.py
│   ├── reporter.py
│   └── target_policy.py
│
├── frontend/
│   ├── index.html
│   └── dashboard.html
│
├── tools/
│   ├── __init__.py
│   ├── nmap_runner.py
│   ├── tshark_runner.py
│   └── traceroute_runner.py
│
├── database/
│   ├── __init__.py
│   └── database.py
│
├── reports/
```

```
│
├── README.md
├── requirements.txt
└── .gitignore
```

8. MAJOR MODULES

8.1 Flask Backend

The Flask backend acts as the central controller of NETFORGE.

It receives requests from the dashboard, executes the required analysis modules, processes their results, and returns structured data to the frontend.

The backend provides APIs for:

- Network scanning
- DNS analysis
- Traceroute
- Traffic capture
- Risk assessment
- Reports
- History
- Analytics

8.2 Nmap Scanner

Nmap is used for network discovery and service identification.

NETFORGE uses an Nmap wrapper to execute authorized scans and collect information such as:

- Open ports
- Protocols
- Services
- Service versions
- Host availability

For example, during testing against the local system, NETFORGE successfully identified:

```
22/tcp open ssh
OpenSSH 9.7p1
```

The Nmap output is then passed to the analyzer.

9. SECURITY ANALYSIS ENGINE

The analyzer evaluates discovered services and assigns security classifications.

Example classification:

Service	Risk
SSH	LOW
FTP	HIGH
Telnet	HIGH
Unknown service	REVIEW

The system also generates recommendations based on the detected service.

For example, insecure services can be flagged for review and hardening.

10. DNS ANALYSIS

The DNS module collects hostname and IP information for the selected target.

During testing against localhost, the system successfully returned:

```
Hostname : localhost
IPv4     : 127.0.0.1
IPv6     : ::1
```

This information helps establish basic network identity and address resolution.

11. TRACEROUTE ANALYSIS

The traceroute module examines the network path between the testing system and the target.

It can provide information about intermediate network hops and connectivity.

This is useful for:

- Network troubleshooting
- Understanding routing
- Identifying path changes
- Diagnosing connectivity problems

12. TRAFFIC ANALYSIS

NETFORGE integrates TShark for network traffic capture and analysis.

The traffic module can collect information such as:

- Packet count
- Protocol information
- Source and destination information
- Traffic characteristics

The captured traffic is then analyzed by NETFORGE's analysis components.

This allows the dashboard to provide a simplified view of network activity.

13. RISK ASSESSMENT ENGINE

One of the major features of NETFORGE is the risk engine.

The engine evaluates collected security information and generates an overall risk score.

Example assessment:

```
Risk Score : 5
Risk Level : LOW
Findings   : 1
Packets    : 25
Traffic    : NORMAL
```

The risk engine categorizes the assessment into levels such as:

- LOW
- MEDIUM
- HIGH
- CRITICAL

This allows users to quickly understand the security condition of the assessed target.

14. TARGET AUTHORIZATION POLICY

NETFORGE includes a target-policy mechanism to encourage responsible security testing.

The platform supports two operating modes:

Local Mode

Designed for:

- localhost
- private IP addresses
- controlled laboratory environments

External Mode

Designed for authorized external targets.

The user must explicitly select the external testing mode and acknowledge authorization before performing an external assessment.

This design helps reinforce the principle that security testing should only be performed against systems for which the tester has permission.

15. WEB DASHBOARD

NETFORGE provides a professional web dashboard for interacting with the platform.

The dashboard provides:

- Target selection
- Scan controls
- Risk overview
- Security findings
- DNS information
- Traceroute results
- Traffic analysis
- Nmap results
- Historical scans
- Generated reports
- Analytics

The dashboard reduces the need to execute every security command manually from the terminal.

16. DATABASE AND HISTORY

NETFORGE uses SQLite to maintain historical information.

The database can store information related to:

- Scans
- Targets
- Risk assessments
- Findings
- Results
- Timestamps

Historical data allows users to compare assessments over time.

17. AUTOMATED REPORTING

NETFORGE can generate HTML security reports.

Example report format:

```
NETFORGE_Report_YYYYMMDD_HHMMSS.html
```

Reports can contain:

- Target information

- Scan information
- Open ports
- Service details
- Security findings
- Risk score
- DNS information
- Network analysis
- Traffic statistics
- Recommendations

This makes the project useful not only for testing but also for documentation.

18. TESTING AND RESULTS

The system was tested in a controlled Kali Linux environment.

Nmap Testing

The scanner successfully identified an SSH service running on localhost.

DNS Testing

The DNS analyzer successfully identified:

- localhost
- 127.0.0.1
- ::1

Traceroute Testing

The traceroute module successfully performed route analysis.

TShark Testing

The traffic module successfully captured packets and generated traffic statistics.

Risk Engine Testing

A sample assessment produced:

```
Risk Score : 5
Risk Level : LOW
Findings   : 1
Packets    : 25
Traffic    : NORMAL
```

Database Testing

Scan history was successfully stored in SQLite.

Report Testing

HTML security reports were successfully generated.

19. ADVANTAGES

NETFORGE provides several advantages:

- Centralized security-testing interface.
- Integration with Kali Linux tools.
- Automated security analysis.
- Risk-based assessment.
- Historical scan storage.
- Automated reporting.
- Beginner-friendly dashboard.
- Modular Python architecture.
- Suitable for cybersecurity laboratories.
- GitHub-ready portfolio project.

20. LIMITATIONS

The current version has some limitations:

11. Results depend on the availability of underlying Kali Linux tools.
12. Some network operations require appropriate system privileges.
13. Traffic capture depends on available network interfaces.
14. Risk scoring is rule-based rather than based on a full vulnerability database.
15. External testing requires proper authorization.
16. The dashboard's Chart.js visualization depends on the required frontend resources being available.

21. FUTURE ENHANCEMENTS

Future versions of NETFORGE could include:

- CVE vulnerability mapping.
- CVSS-based vulnerability scoring.
- Automated vulnerability database integration.
- PDF report generation.
- User authentication.
- Role-based access control.
- Advanced packet analytics.
- Real-time traffic monitoring.
- Scheduled security assessments.

- Email report notifications.
- Asset inventory.
- Multi-target assessment.
- Docker deployment.
- Improved authorization controls.
- Security trend dashboards.
- Export to CSV and JSON.

22. SECURITY AND ETHICAL CONSIDERATIONS

NETFORGE is intended for authorized security testing only.

Users should only scan:

- Their own systems.
- Their own networks.
- Dedicated cybersecurity laboratories.
- Systems for which they have explicit permission.

The project incorporates target-policy controls to encourage responsible use.

The platform should not be used for unauthorized reconnaissance, monitoring, or testing of third-party systems.

23. GITHUB INTEGRATION

The project has been uploaded to GitHub and uses Git for version control.

Repository: NETFORGE GitHub Repository

The repository contains:

- Source code
- README documentation
- Configuration files
- Requirements
- Project structure
- Security controls

Sensitive information such as private keys, databases, generated reports, and environment files are excluded through .gitignore.

24. CONCLUSION

NETFORGE demonstrates the development of a practical network-security assessment platform using Python, Flask, Kali Linux security tools, SQLite, and a web-based dashboard.

The project brings together network scanning, DNS analysis, traceroute, traffic monitoring, security analysis, risk scoring, historical data storage, and automated reporting into a single platform.

Through this project, practical knowledge was developed in areas including network security, Linux, Python programming, Flask web development, security-tool integration, database management, traffic analysis, risk assessment, and Git/GitHub-based project management.

NETFORGE can serve as a strong cybersecurity portfolio project because it demonstrates not only knowledge of individual security tools, but also the ability to integrate those tools into a complete working application.